



Contactless Smart Bin

Parth Atulkumar Shah

This smart bin can help maintain clean and hygienic environment as garbage can be thrown into it without touching it. The bin has a motor attached to its lid and uses an HC-SR04 sensor to detect hands approaching it. The bin could be useful in crowded places like malls, offices, schools, colleges, and hospitals.

Most Indian cities, towns, and villages are not well equipped for garbage collection. Even when the dust-bins start overflowing, no one clears them. This smart bin could be a good solution in such situations as it will send an alert when the bin becomes full.

The bin device has following features:

1. The bin opens automatically when someone approaches it, thus keeping the process of disposal non-contact and hygienic.

2. The bin can alert the garbage

collectors when it needs to be emptied. Thus, municipal corporations can have the garbage collected well in time, before it starts overflowing, and thus prevent their ugly spillage.

3. Hospitals can use the smart bins to keep the cities free of bio-waste that could potentially be hazardous and spread infections.

Circuit and working

The author's prototype is shown in Fig. 1 and circuit diagram of the smart bin is shown in Fig. 2. It is built around Arduino Uno Board1, HC-SR04 ultrasonic sensor US1, and servo motor M1.

When the ultrasonic sensor registers hand movement in front of the bin, it generates a signal

at its Echo and Trigger pins. This signal is received and processed by the controller (Arduino) and sent to its digital output pins. The signal goes to servo motor MG90S to rotate the bin's lid attached to it by required degrees. The degree of rotation, decided by size

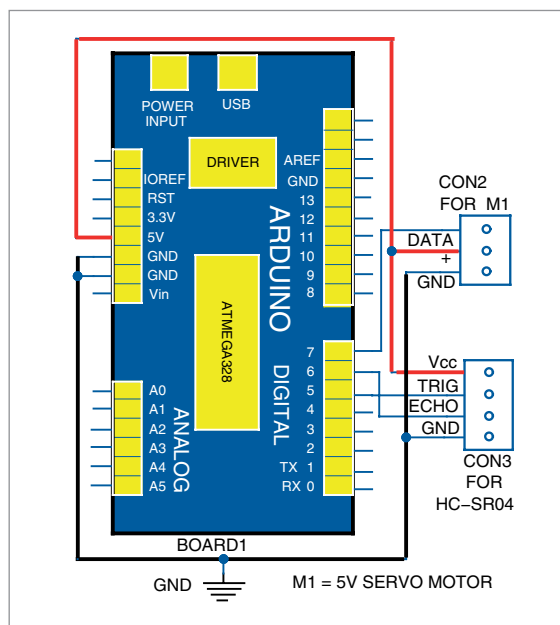


Fig. 2: Circuit diagram



Fig. 1: Author's prototype



Fig. 3: Servo motor installed in the lid